Water quality (DP1.20288.001)

Measurement

Water quality measurements include specific conductance, dissolved oxygen, pH, chlorophyll a (at lake and river sites only), turbidity and fluorescent dissolved organic matter (fDOM).

Collection methodology

Measured using YSI EXO2 Multisondes located at both the S1 (upstream) and S2 (downstream) locations of wadeable streams and on the buoy at lakes and river sites. At wadeable stream sites measurements are collected every minute. At lakes and river sites measurements are collected at 0.5 m below the surface every 5 minutes, and a vertical profile of the water column is collected every 4 hours.

fDOM is corrected for effects of temperature and absorbance on measured fluorescence. Absorbance data are obtained from a co-located SUNA. For this reason, fDOM in wadeable stream sites is only reported for the S2 (downstream) location.

For information about disturbances, land management activities, and other incidents that may impact data at NEON sites, see the Site management and event reporting (DP1.10111.001) data product.

Maintenance and calibration

The multisonde has an antifouling copper guard and an automated wiper which cleans the optical lens before every fifth measurement. Manual cleaning is performed bi-weekly and field calibration is performed monthly. Sensors are returned to the NEON CALVAL lab annually for refresh.

The National Ecological Observatory Network (NEON) is a major facility fully funded by the National Science Foundation and operated by Battelle.



Multisonde inside protective PVC housing.

Data package contents

ais_maintenance: Information related to aquatic sensor and infrastructure maintenance
ais_multisondeCleanCal: Information related to multisonde cleaning and calibration activities
waq_instantaneous: Instantaneous water quality measurements
variables: Description and units for each column of data in data tables
readme: Data product description, issue log, and other metadata about the data product
sensor_positions: Geospatial locations of individual sensors

Data quality

Each measurement is accompanied by a final quality flag ("...FinalQF"). Data with a final quality flag of 1 are potentially inaccurate and should only be used with caution. The final quality flag is based on automated QA /QC tests, including range, gap, and spike tests, as well as a manually set science review flag if applicable ("...

FinalQFSciRvw"). This data product also has automated flags for sensor lamp temperature and internal humidity. Each measurement is accompanied by an estimate of measurement uncertainty, expressed at the 95% confidence level ("...ExpUncert"), which comprises known and quantifiable uncertainties.

Please note that quality checks are comprehensive but not exhaustive; therefore, unknown data quality issues may exist. Also note that, conversely, some quality-flagged data are still usable depending on the scientific use case. Additionally, these Level 1 data are not currently gap-filled or corrected for sensor drift or shifts, such as those introduced by sensor swaps or field calibrations. Users are advised to evaluate quality of the data as relevant to the scientific research question being addressed, perform data review and post-processing prior to analysis, and use the data quality flags, issue logs, and maintenance records included in download packages to aid interpretation. A tutorial with examples of how to do this can be found here.

Standard calculations

For wrapper functions to download data from the API, and functions to merge tabular data files across sites and months, NEON provides the neonUtilities package in R and the neonutilities package in Python. See the Download and Explore NEON Data tutorial for introductory instructions in both programming languages.

High frequency (1-min) data collected at wadeable stream sites can be averaged to reduce sensor noise and match the temporal resolution of other NEON data products. Pay careful attention to missing and quality flagged measurements and omit from the average as appropriate. Sensor height (zOffset; m), the latitude, longitude (referenceLatitude, referenceLongitude; °), and elevation (m) of the in stream infrastructure base are in the sensor positions file (...sensor_positions...csv). Use the HOR.VER component of the time series file name (horizontalPosition and verticalPosition if stacked using neonUtilities) to link to the corresponding row in the HOR.VER column of the sensor positions file. An HOR of 101 or 111 correspond to Sensor Set 1 (upstream) and HOR of 102 or 112 correspond to Sensor Set 2 (downstream). An HOR of 103 correspond to buoys.

Documentation

NEON Aquatic Sampling Strategy
NEON.DOC.001152vD | 846.5 KiB | PDF

NEON Sensor Command, Control and Configuration (C3) Document: Multisonde, Stream
NEON.DOC.001166vJ | 462 KiB | PDF

NEON Preventive Maintenance Procedure: AIS Surface Water Quality Multisonde
NEON.DOC.001569vE | 12.8 MiB | PDF

NEON Sensor Command, Control and Configuration (C3) Document: Buoy Meteorological Station and Submerged Sensor Assembly
NEON.DOC.003808vP | 1.8 MiB | PDF

NEON Algorithm Theoretical Basis Document (ATBD): Water Quality
NEON.DOC.004931vC | 1.6 MiB | PDF

NEON User Guide to AIS Maintenance Data
NEON_ISmaintenance_vB | 217.8 KiB | PDF

For more information on data product documentation, see:
https://data.neonscience.org/data-products/DP1.20288.001

Citation

To cite data from value: "Water quality" (DP1.20288.001), see citation here:
https://data.neonscience.org/data-products/DP1.20288.001
For general guidance in citing NEON data and documentation, see the citation guidelines page:
https://www.neonscience.org/data-samples/guidelines-policies/citing

Contact Us

NEON welcomes discussion with data users! Reach out with any questions or concerns about NEON data:
Contact Us